

Diode Threshold Priority - LED Selector Circuit when motor turn

How It Works

This circuit illustrates how parallel branches with different forward voltage thresholds compete for current when sharing a common current-limiting resistor.

One branch contains a single green LED ($V_f \approx 2.2 \text{ V}$).

The other branch consists of three series silicon diodes ($\approx 0.7 \text{ V}$ each) + one red LED ($V_f \approx 1.9 \text{ V}$), creating a much higher total threshold ($\approx 4.0 \text{ V}$).

With the $1 \text{ k}\Omega$ resistor in the common return path, only the green LED lights: it turns on first at the lower voltage, clamps the voltage across both parallel branches to $\sim 2.2 \text{ V}$, and prevents the higher-threshold red branch from conducting (Kirchhoff's voltage law in parallel paths). The resistor drops the remaining $\sim 9.8 \text{ V}$ and limits current to a safe $\sim 10 \text{ mA}$.

When the resistor is removed (or shorted), both branches receive the full 12 V , so both LEDs illuminate brightly (but without current limiting they risk damage).

This shows why the path with the lowest forward voltage drop always wins priority in parallel arrangements – a key concept in diode logic, steering, and threshold-based selection circuits.





